

Physics

Title: UNIT 1 - MECHANICS & PROPERTIES OF MATTER

- **Newton's Three Laws of Motion:**
 1. **Law of Inertia:** Objects resist changes in their state of motion.
 2. **Fundamental Law ($F = ma$):** Acceleration is produced when a force acts on a mass.
 3. **Action and Reaction:** Every action has an equal and opposite reaction.
- **Work, Power, and Energy:**
 - *Work:* Done when a force moves an object.
 - *Kinetic Energy:* Energy of an object in motion ($K.E. = \frac{1}{2}mv^2$).
 - *Potential Energy:* Stored energy based on position ($P.E. = mgh$).
- **Thermodynamics:** The study of heat transfer and its transformation into work.

Newton's Second Law

